

复流形上的几何 III

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1 Lecture 4

Recall: Set up: Let

$$(E, v) : 0 \rightarrow E_0 \xrightarrow{v} E_1 \xrightarrow{v} \cdots \xrightarrow{v} E_n \rightarrow 0$$

where

1. E_j are holomorphic vector bundles on a connected complex manifold X .
2. v are holomorphic bundle maps ($\bar{\partial}^E \cdot v = [\bar{\partial}^E, v] = 0$) and $v^2 = 0$.

Let $h^E = \bigoplus h^{E_j}$ be a Hermitian metric on E , v^* the adjoint of v w.r.t. h^E , and $V = v + v^*$.

For $u > 0$, define the superconnection $A_u = \nabla^E + \sqrt{u}V$.

Last time we have prove the following theorem:

Theorem 1.1.

$$\frac{\partial}{\partial u} \text{Tr}_s [\exp(-A_u^2)] = -\frac{1}{\sqrt{u}} \bar{\partial} \partial \text{Tr}_s [N \exp(-A_u^2)]$$

For $x \in X$, let

$$H_x^i := H_x^i(E) = \frac{\text{Ker } v|_{E_{i,x}}}{\text{Im } v|_{E_{i-1,x}}}$$

Assume that $\dim H_x^i$ are constant $\forall i$, then

Proposition 1.2. H_x^i form a holomorphic vector bundle H^i on X .

Proof. ommitted. □